

Technology Stack

Date	2 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML,CSS,JavaScript	Used to create a simple, responsive, and user-friendly interface for entering crop prediction inputs and displaying results.

2	Backend / Server-Side	Python (Flask)	Flask is lightweight, easy to develop, and integrates well with Machine Learning models for handling prediction requests.
3	Database / Data Storage	CSV Dataset (Crop Recommendation Dataset)	Stores agricultural data used to train the Machine Learning model. A database is not required for the current version.
4	Cloud / Hosting / Deployment	Local Host (Flask) <i>(or Render, if deployed)</i>	Used to run and test the application during development. Can be deployed to cloud platforms in the future.

5	Version Control & CI/CD	Git & GitHub	Used for version control, source code management, backup, and collaboration.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Visual Studio Code	Scikit-learn is used for the Machine Learning model, Pandas and NumPy for data processing, Joblib for loading the trained model, and VS Code for development and debugging.